VISHESH KUMAR

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CURRENT ROLE

PhD Scholar December 2022 - present
Department of Data Science and Engineering
IISER Bhopal,Bhopal M.P., India

EDUCATION

M.Sc. (Applied Mathematics) July 2019 - August 2021
Indian Institute of Engineering Science and Technology (IIEST), Shibpur, India. CGPA: 8.8
B.Sc. Physical Science July 2016 - June 2019
Motilal Nehru College, University of Delhi, India. CGPA: 7.712
HSC, Class XII March 2016
Bihar School Examination Board, Gaura Kunwar Inter College, Adampur, Bihar, India. Percentage: 71.4 %
SSC, Class X March 2014
Bihar School Examination Board, Patna, Bihar, India. Percentage: 66.6 %

RESEARCH INTERESTS

AI Security, Trustworthy AI, Computer Vision, ML/DL, Presentation Attacks, Adversarial Robustness, Deepfakes, Bias, Frame Theory

FELLOWSHIP AND AWARDS

- Visvesvaraya PhD Fellowship, Government of India December 2022 - Present
- Secured Rank-1 in the Second year of B.Sc.
- National Level Exam - IIT JAM (2019,2020), and GATE (2022) Qualified

RESEARCH EXPERIENCE

- Junior Research Fellow on the project **"Dynamical Sampling and Representation of Frames via Iterated Operator System"** [**Code: SP/20/SERB/NKS/DSRFIO**] under the guidance of Prof. Nabin Kumar Sahu at DA-IICT, Gandhinagar, Gujarat August 2022 to December 2022

RESEARCH PUBLICATIONS

Published

1. **V. Kumar**, A. Agarwal, A Unified, Resilient, and Explainable Adversarial Patch Detector, IEEE/CVF, CVPR, 2025
2. **V. Kumar**, A. Agarwal, Are Object Recognition Models Effective and Unbiased for Biometric Recognition?, IEEE International Joint Conference on Biometrics (IJCB), 2024
3. **V. Kumar**, A. Agarwal, On Unconstrained Ear Recognition For Privacy-Preserving Authentication, Workshop on Advances of Mobile and Wearable Biometrics (WAMWB), in conjunction with ACM International Conference on Mobile Human-Computer Interaction (MobileHCI), 2023

Under Peer Review

1. **V. Kumar**, S. Shukla, A. Agarwal, Robustness Benchmarking of Convolutional and Transformer Architectures for Image Classification, IEEE Transactions on Big Data, 2025 *(Review Submitted)*
2. **V. Kumar**, A. Agarwal, Fortifying Deep Networks: How Training Data Shapes Robustness to Corruption, 2025
3. **V. Kumar**, A. Agarwal, The Unseen Adversaries: Robust and Generalized Defense Against Adversarial Patches, SSRN, 2024

   .
4. NK Sahu, **V Rajput**. Dynamical Representation of Frames in Tensor Product of Hardy Spaces. arXiv preprint arXiv:2308.11330 (2023)

PROFESIONAL SITES

- Linkedin https://www.linkedin.com/in/vishesh-rajput-7a277119b/
- Google Scholar vishesh_google_scholar

RESEARCH ABSTRACT

Artificial intelligence (AI) is increasingly used in critical applications such as autonomous vehicles, robotics, smart home and city technologies, industrial automation, video surveillance, healthcare, and facial recognition systems. These AI frameworks often incorporate Deep Neural Network (DNN) models, which are susceptible to physical adversarial attacks. In such attacks, attackers place a patterned sub-image over real-world objects to deceive AI systems. For example, the adversarial patch attack can not only provide illegal access to intruders but, in a similar fashion, can make humans invisible to human detectors, highlighting a significant security vulnerability.

Researchers are constantly developing new defenses or protection strategies to tackle the limitations of DNNs against physical adversarial patches, but understanding their decision-making processes is increasingly important, and understanding how these complex models reach decisions remains a challenge. As attackers develop new adversarial patches to exploit vulnerabilities in AI systems, defense mechanisms that are designed to counteract known threats become insufficient. These defenses are typically built on datasets of known patch attacks and use predetermined algorithms to detect and mitigate these threats. However, they lack the capability to adapt when attackers modify their approaches to create adversarial patches or when entirely new types of adversarial patches are deployed. The goal is to create systems that are more robust, generalized, and explainable, not only reacting to known threats but also anticipating and mitigating new adversarial patch threats as they emerge.